

MIKOLAJKI, Poland

WMO: 122800

Lat: 53.7892N

Lon: 21.5886E

Elev: 421

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	6.7	-2.8	4.8	1.6	2.4	6.3	7.8	21.8	18.0	20.1	21.8	4.9	340	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	82.8	68.5	79.7	66.4	76.8	64.9	70.4	79.5	68.6	76.7	66.8	73.9	6.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	102.1	74.7	65.6	96.1	72.5	64.0	90.6	70.7	34.6	79.4	33.0	77.2	31.6	73.9	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.9	14.0	-3.2	88.2	6.1	3.0	-7.6	90.4	-11.1	92.2	-14.5	93.9	-18.9	96.1
			-3.7	73.5	5.9	1.9	-7.9	74.9	-11.4	76.0	-14.7	77.1	-18.9	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	26.8	28.7	35.1	46.1	55.5	61.5	65.8	65.1	56.8	46.8	38.2	30.7
	DBStd	15.68	10.16	9.62	7.84	7.57	6.99	5.89	5.45	5.28	5.74	6.70	7.11	9.15
	HDD50	3074	720	597	464	161	26	1	0	0	10	142	356	597
	HDD65	6941	1185	1017	928	566	305	137	56	65	253	563	804	1062
	CDD50	1803	0	0	1	45	195	346	491	467	213	44	1	0
	CDD65	196	0	0	0	1	9	32	82	67	5	0	0	0
	CDH74	1299	0	0	0	13	79	235	533	408	31	0	0	0
	CDH80	245	0	0	0	1	3	39	118	80	4	0	0	0
Wind	WSAvg	6.9	8.1	7.8	7.6	6.9	6.1	5.9	5.5	5.3	5.8	7.1	8.3	8.3
Precipitation	PrecAvg	23.9	1.3	1.2	1.4	1.4	2.3	2.8	3.1	3.1	2.2	1.9	1.8	1.4
	PrecMax	29.4	3.4	2.4	3.5	3.5	6.4	4.7	6.9	9.0	5.1	5.1	3.5	2.6
	PrecMin	17.3	0.4	0.4	0.2	0.1	1.2	0.9	0.6	0.7	0.7	0.0	0.5	0.3
	PrecStd	3.3	0.7	0.5	0.7	0.8	1.0	1.1	1.3	1.7	1.3	1.3	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	48.3	59.2	75.3	80.1	85.1	87.5	86.6	78.4	66.6	54.8	49.4
		MCWB	44.3	43.9	49.2	59.3	64.6	68.7	69.9	70.6	66.1	58.5	51.5	47.1
	2%	DB	43.4	45.4	53.5	68.6	76.4	79.9	83.6	81.9	72.7	62.2	51.6	46.3
		MCWB	41.6	42.7	45.7	55.2	62.8	66.0	69.2	67.9	62.2	56.6	49.0	44.1
	5%	DB	40.9	42.7	49.2	63.7	72.6	76.4	79.9	78.3	69.2	59.0	49.0	43.8
		MCWB	39.4	40.3	43.6	52.3	60.3	64.1	67.6	65.6	60.5	54.7	46.9	42.2
	10%	DB	38.3	40.2	45.7	59.2	68.7	73.0	76.5	75.3	66.1	56.2	47.0	41.3
		MCWB	37.1	38.2	41.7	49.9	58.4	62.3	65.9	64.3	59.0	53.1	45.3	39.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	45.7	50.4	60.7	66.9	71.2	73.3	72.6	66.7	60.2	52.5	47.7
		MCDB	46.4	46.9	56.9	74.8	76.9	82.1	82.1	82.3	74.8	64.6	54.2	49.0
	2%	WB	42.0	42.8	47.0	56.4	64.2	68.1	70.8	69.9	64.0	57.5	49.5	44.7
		MCDB	43.3	44.8	51.9	65.5	73.8	76.2	79.6	79.1	70.4	60.9	51.1	45.9
	5%	WB	39.4	40.6	44.3	53.4	61.9	65.9	69.1	67.9	61.9	55.4	47.3	42.2
		MCDB	40.8	42.5	48.0	61.6	70.3	72.9	77.5	75.3	67.2	58.3	48.6	43.5
	10%	WB	37.2	38.1	41.9	50.8	59.5	63.7	67.2	65.9	59.9	53.4	45.5	40.1
		MCDB	38.2	40.0	45.4	57.7	66.9	71.0	74.5	72.5	64.8	56.1	46.9	41.3
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.2	12.4	16.4	17.3	15.9	15.5	15.5	13.3	10.6	6.8	6.6
		MCDBR	7.6	10.0	17.8	23.7	22.1	21.0	20.0	20.0	17.8	14.2	9.0	7.4
		MCWBR	7.0	7.9	11.6	12.4	10.9	10.0	8.6	8.8	9.0	9.1	7.3	6.8
	5% WB	MCDBR	7.5	9.7	16.5	20.9	20.2	18.9	18.0	17.8	16.0	13.0	8.3	7.4
MCWBR		7.0	8.2	11.6	11.9	11.0	10.3	8.9	8.9	8.8	8.9	7.2	6.9	
Clear-Sky Solar Irradiance	taub	0.301	0.331	0.367	0.410	0.399	0.396	0.429	0.421	0.380	0.357	0.343	0.300	
	taud	2.218	2.217	2.239	2.198	2.290	2.326	2.250	2.270	2.359	2.406	2.300	2.253	
	Ebn at Noon	205	237	255	259	269	271	258	252	249	229	187	180	
	Edh at Noon	20	28	34	40	39	38	40	37	30	23	19	16	
All-Sky Solar Radiation	RadAvg	195	420	813	1308	1680	1750	1619	1390	986	515	194	130	
	RadStd	34	71	130	171	162	154	176	131	129	89	22	15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	N/A
Regional (18 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-268	+135	N/A

Nomenclature: See separate page